Jack Dubie

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EXPERIENCE

Frontier Machines — CTO · 2023–2026

- Deployed an economically viable fleet of pouch-packing robots running π0.7 with an in-house-developed supervisor model at 97.6% autonomy, packing 45,000 pouches to date; featured by Physical Intelligence in its API beta announcement.
- Built a remote teleop system for dexterous manipulation on Meta Quest 3S, I2RT YAM Pro arms, and an NVIDIA Jetson Orin with GMSL2 cameras, using FEC and HEVC hardware encode/decode, achieving 85 ms glass-to-glass latency with operators 2,000 miles away.
- Fine-tuned a π0.5 policy for quasi-direct-drive YAMs entirely from Meta Quest 3S UMI data by building an embodiment pipeline that retargets demos with offline iterative learning control, solving a demo task autonomously with zero robot-collected fine-tuning samples.
- Implemented training-time RTC in π0.5 for a jam-prone task, conditioning on committed action prefixes to smooth asynchronous chunk execution, cutting peak (p99) jerk 4–5x and making the task viable.
- Doubled task throughput on a bimanual parallel-gripper robot by designing task-specific fingers and 3D-printed jigs, which also gave the wrist camera a much clearer view of the task.
- Designed and built a new UMI data-collection rig, hardware and software, replacing the exoskeleton with Meta Quest 3S tracking and far better operator kinematic feedback, cutting BOM 94% ($8k → $500) and tracking error 20% while matching the existing rig's policy performance.
- Reduced YAM position tracking error 87% (30 mm → 4 mm) so position-control policies could run on compliant quasi-direct-drive arms, using an inverse-dynamics model and gain tuning after evaluating MPC and action prediction.
- Calibrated exoskeleton tracking against OptiTrack ground truth with quasi-Newton optimization over link lengths and encoder offsets, cutting tracking error 85% (40 mm → 6 mm).
- Evaluated UMI variants end to end, from collection to policy rollout, across xArm 6 and I2RT YAM arms with EMF, OptiTrack, Vive Tracker 3.0, and ORB-SLAM tracking.

Independent Exploration — Full-Stack Robotics Engineer · 2023–2024

- Showed low-cost home robots can be run remotely: built a <$1k BOM toy cleanup robot, advised by Tony Zhao, pairing leader-follower arms with an internet teleop link, and demonstrated it picking up toys while teleoperated from 7,000 miles away.

Ladder — Co-founder · 2015–2023

- Built ML training and inference for instant life-insurance underwriting across ~1M applications, predicting mortality from prescription fills, driving history, and credit history, providing $113B in coverage to families.
- Shipped continuous integration, a fast in-memory end-to-end testing framework, and a suite of APIs and SDKs powering deep co-branded partner integrations.

Dropbox — Software Engineer · 2013–2015

- Developed Auto-swipe for Mailbox, which learns from each user's archive and snooze swipes to triage matching future email before it's opened.
- Co-inventor on 8 granted US patents across 3 patent families: dynamic email filter generation, message thread identification, and shared-folder-backed workspaces.

EDUCATION

Stanford University · MS Computer Science '13 · BS Mathematics '11 · BS Computer Science '11